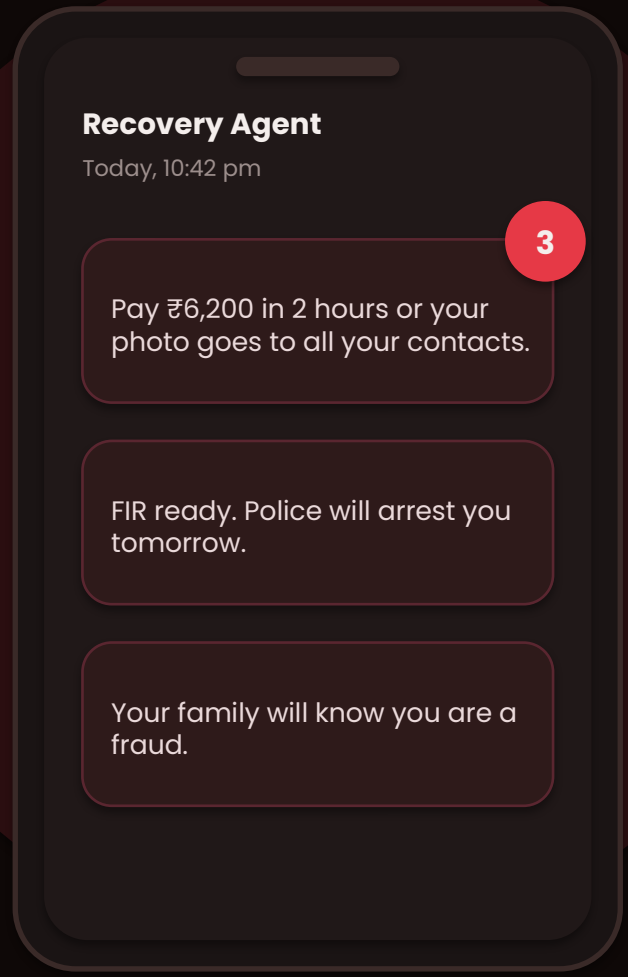
SOCIAL & SUSTAINABILITY PROJECT

Digital Loan App Harassment in India

Framing the crisis through the UN SDGs, and our solution: LoanShield.



1 8 10 16 Four SDGs, one problem

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THE CRISIS

Illegal digital lending traps borrowers

1 Quick loans attract the vulnerable

Instant approval, no paperwork, aimed at people banks turn away.

2 Hidden fees deepen debt

Large cuts before the money arrives, then short deadlines and penalties.

3 Aggressive recovery tactics

Threats, blackmail and shaming when the borrower cannot pay.

The harm spreads: families, workplaces and whole communities lose trust in digital finance.

WHY IT MATTERS

Harassment tactics & real-world harm

Threatening calls & messages

Abusive language and fake legal notices create constant fear.

Contact-list blackmail

Personal data is weaponised to shame and pressure borrowers.

Fear, stress & family conflict

Repeated harassment causes psychological harm and tears families apart.

Privacy violations

Data misuse exposes victims and destroys trust in digital systems.

ACCESS WITHOUT PROTECTION

NO INSTITUTIONAL ACCOUNTABILITY

UNSUSTAINABLE DIGITAL FINANCE

1

SDG 1

No Poverty

Predatory lending creates a poverty trap

1 **High charges**

Push vulnerable households deeper into debt.

2 **Debt cycles**

Forced to re-borrow just to survive.

3 **Essential cuts**

Less spending on food, health and education.

4 **Financial ruin**

Poverty becomes impossible to escape.

SDG 1: predatory lending does not just cause debt. It makes poverty structurally harder to escape.

Unsafe lending undermines fair economic participation

8

SDG 8
Decent Work & Economic Growth

Livelihoods at risk

Harassment calls at work disrupt employment. Constant stress reduces productivity. Predatory debt makes decent work meaningless.

10

SDG 10
Reduced Inequalities

Inequality widens

Vulnerable and marginalised groups face the highest risk of exploitation. Without safe credit, existing inequalities only grow.

16

SDG 16

Peace, Justice & Strong Institutions

Threats, data abuse & eroding trust

Threats & fake legal notices

Create constant fear and insecurity, undermining personal safety and dignity.

Privacy violations

Misuse of personal data weakens trust in digital institutions and financial systems.

No accessible justice

Victims lack clear pathways for accountability, leaving exploiters unchallenged.

Without strong institutions to regulate lending and protect data, digital finance becomes a tool of coercion.

ONE PROBLEM, FOUR SDGs

Digital finance must empower, not exploit.

1

Deepens poverty & debt traps

8

Disrupts livelihoods & work

10

Widens existing inequalities

16

Erodes trust & justice

Loan-app harassment is not just a fintech issue. It is a sustainable development crisis. Safe, fair and accountable digital finance is essential for inclusive growth in India.

PROBLEM RECAP

Where the problem leaves victims

1 Easy loan, hidden cost

₹5,000 promised, ₹3,500 paid, full repayment in 7 days.

2 Threats & blackmail

Contact lists, morphed photos and fake police notices.

3 Silence & rebrand

The victim pays and stays quiet. The app returns under a new name.

What is missing: a way for victims to fight back, together, with proof.

1 8 10 16

Our solution answers each stage of it.

OBJECTIVES

What our solution sets out to do

1 Calm the victim

Answer every threat with the facts and the law it breaks.

2 Build evidence

Seal each message as tamper-proof proof.

3 Protect the contacts

Warn family and friends before the scammers reach them.

4 Unmask the operator

Link rebranded apps by UPI ID, phone number and wording.

5 Offer a fair alternative

Small, fair loans through SHGs and microfinance groups.

6 Reach everyone

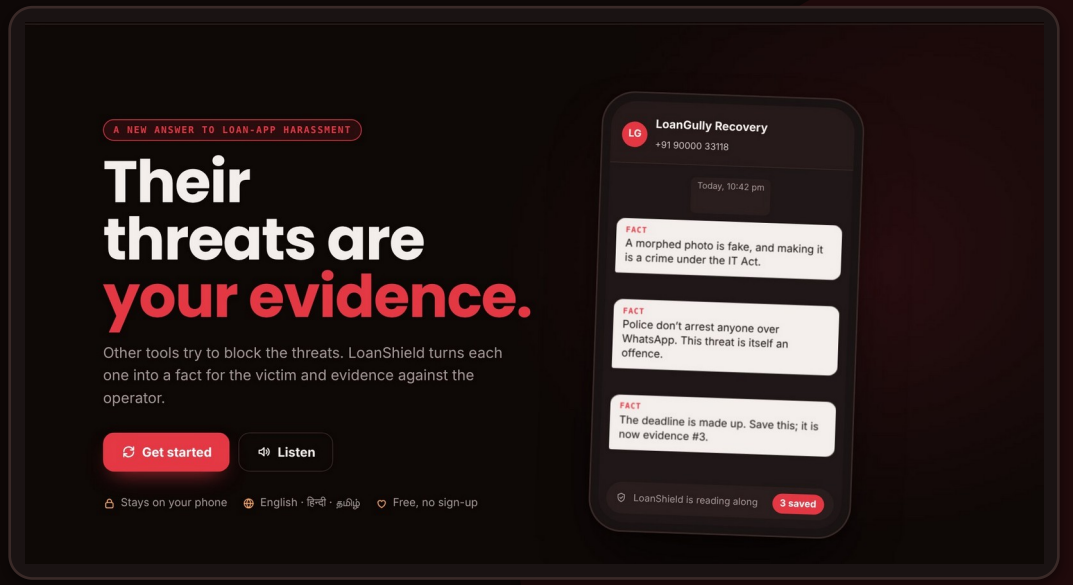
Tamil, Hindi and English; help for people who can’t use phones.

OUR SOLUTION

LoanShield

Their threats become evidence.

A community system with a website at its centre. Other tools try to block the threats. LoanShield turns each one into a fact for the victim and evidence against the operator.



Flip Every threat is answered with what is actually true.

Seal Every message becomes tamper-proof evidence.

Link Every UPI ID leads back to one operator.

Live https://karthik705.github.io/loanshield/

HOW IT WORKS

Five stages, one loop

1 Prevent

COMMUNITY

- Real-cost camps at SHGs and colleges
- Fair small loans via SHGs

→

2 Detect

WEBSITE

- Links reports by UPI ID and number
- Public only after 3 people report

→

3 Protect

WEBSITE + NGO

- Flips each threat into fact and law
- Warns contacts in 3 languages

→

4 Respond

COMMUNITY

- Legal-aid clinics help file
- Leaked-photo takedowns by NGOs

→

5 Reform

COMMUNITY

- Public dashboard of operators
- Reports to app stores

↻ **Loop:** hotspots found in Reform decide where the next Prevent camps run.

● Carried by the website ● Carried by people and partners

THE WEBSITE: ONE PART OF THE SYSTEM

Paste a threat. Watch it flip.

Speak, paste or screenshot

Say what happened; the key details are pulled out.

Fear → Fact

Each threat card turns over to show the truth and the law.

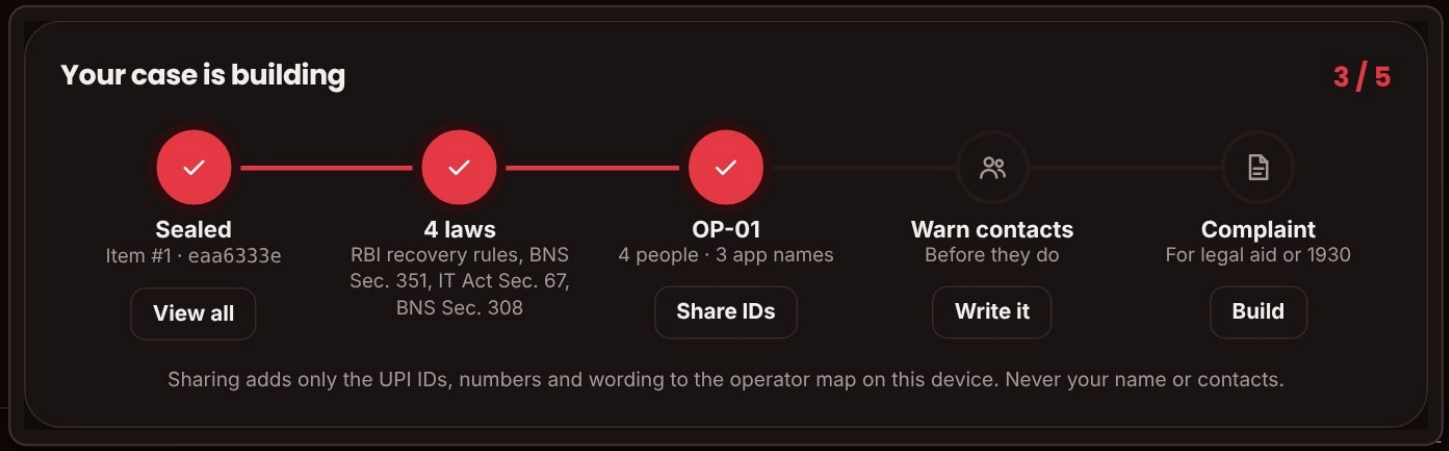
Protect me in one tap

Seals it, links the operator, writes the warning and complaint.

Case tracker

Evidence, laws, operator, warning and complaint in one view.





TECH STACK

Simple tools, chosen to reach everyone

Website

Opens in any phone browser. No app to install.

HTML · CSS · JavaScript

Reading threats

Spots threat words in English, Hindi and Tamil.

Keyword detection + AI for screenshots

Voice

Speak instead of typing; hear the facts read out.

Phone's built-in speech tools

Evidence seal

A digital fingerprint shows if anything changed.

SHA-256 hashing

Linking operators

Matches UPI IDs and numbers across reports.

Pattern matching

Privacy & hosting

Checked on the phone; shared only by choice.

Free hosting on GitHub Pages

FOR PEOPLE WHO STRUGGLE WITH TECH

Two ways in: do it yourself, or get assistance

Do it yourself

Paste, speak or upload a screenshot. About a minute.

Someone is with me

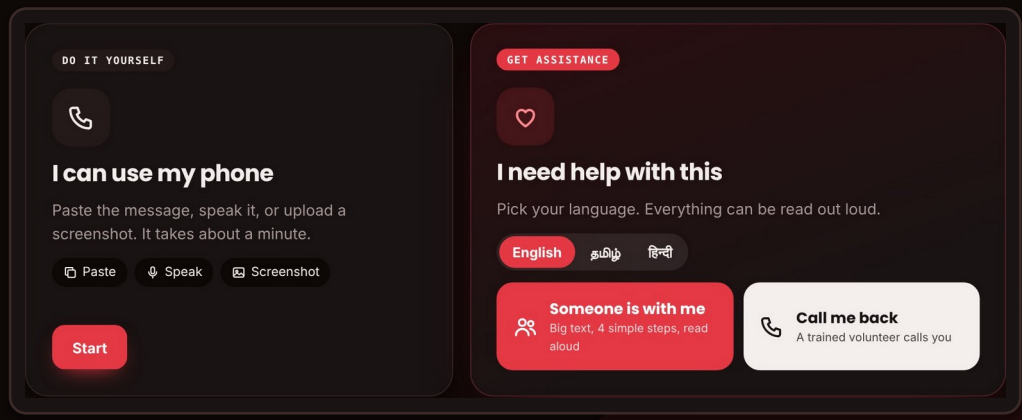
Helper mode: big text, 4 simple steps, read aloud.

Call me back

A trained NGO volunteer calls in their language.

Their language, out loud

Tamil, Hindi or English; facts are spoken.



No smartphone at all? Student volunteers run weekly help desks at SHG meetings and do every step for them.

WHEN A MORPHED PHOTO LEAKS

Evidence is not enough. Dignity comes first.

Before it spreads

- Contact Shield warns family first: “any photo of me is fake”
- The photo arrives already discredited

Within hours

- Ready takedown request for each platform
- Platforms must act within 2 hours of a complaint (IT Rules, 2026)
- StopNCII blocks re-uploads; the image never leaves the phone

For the person

- An NGO counsellor does the takedowns
- A woman volunteer on request
- They never have to see it again

On the website: “This photo is fake. Making it is the crime, not being in it.” One button opens the whole checklist.

NOVELTY

What is new in our approach

USUAL APPROACH	LOANSHIELD
Spam blockers hide the message. The victim is still scared.	**Flips fear into facts** — Each threat is answered with the truth and the law it breaks.
More threats mean more fear. Each victim complains alone.	**Every threat builds the case** — Messages are sealed as evidence; contacts are warned first.
One app is reported. It returns next week with a new name.	**Follows the money** — Reports sharing a UPI ID or number join into one operator.

Other tools defend one phone. LoanShield turns every victim's report into one case against the operator.

SDG IMPACT

How the solution serves the SDGs

10

MAJOR SDG · TARGET 10.2

Reduced Inequalities

These apps exploit the people left behind, financially and digitally. LoanShield serves them first.

- Assisted help: a helper, a call-back or an SHG help desk
- Their language, read aloud: Tamil, Hindi, English
- No app, no sign-up: works on any cheap phone

1

SDG 1 · No Poverty

Fair SHG loans and real-cost checks break the debt trap.

16

SDG 16 · Peace & Justice

Sealed evidence and legal-aid help give victims access to justice.

8

SDG 8 · Decent Work

Contact warnings keep threats away from employers.

FUNDING & PARTNERS

Run with cyber-safety NGOs

CyberPeace Foundation

Cyber-safety NGO. Pilot funding, counsellor network and victim helpdesk.

Responsible Netism

Online-safety NGO. Awareness workshops in colleges and communities.

Cyber Society of India

Chennai-based cyber awareness body. Volunteer training for our pilot.

CSR grants: from responsible fintech companies and banks

Law-college clinics: pro-bono legal help for filing

SHGs & microfinance groups: the fair-loan alternative

Student volunteers: NSS and social clubs run camps

Partners we plan to approach for the pilot. Running cost is low: the website is free to host.

CHALLENGES & LIMITATIONS

What makes it hard, and what it can’t do yet

CHALLENGES → HOW WE HANDLE THEM

Apps rebrand in weeks
→ link by UPI ID and number, not app name

Shame stops reporting
→ private by default, NGO counsellors

False or spiteful reports
→ public only after 3 different people

Low digital literacy
→ assisted help, voice and local languages

LIMITATIONS

Keyword scanner can miss new wording

Tested on 50 team-written messages, not field data

Evidence is stored only on the victim’s own device

Legal mapping is a guide, not legal advice

People-side stages depend on NGO partners

FUTURE SCOPE

Where LoanShield goes next

1 Android app

Share any message straight in; log late-night calls automatically.

2 Smarter reading

A trained model for Tamil, Telugu and Bengali threats.

3 WhatsApp bot

Forward a threat to one number; get the facts back as a voice note.

4 Voice notes

Turn threatening voice messages into text evidence.

5 90-day pilot

One SHG federation and one cyber-safety NGO in Chennai.

6 Open data for NGOs

Share operator links with partners to speed up takedowns.

THANK YOU

Safer loans.
Stronger people.

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Try LoanShield

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